

temp lab01-02

Positional notation

Positional notation

- Usually means the extension to any base (2, ..., 8,10,16 etc.) of the Hindu–Arabic numeral system (or decimal system).
- A positional system is a numeral system in which the contribution of a digit to the value of a number is the value of the digit multiplied by a factor determined by the position of the digit.
- In modern positional systems, such as the decimal system, the position of the digit means that its value must be multiplied by some value: in 444, the three identical symbols represent four hundreds, four tens, and four units, respectively, due to their different positions in the digit string.

Base of the numeral system base

- In mathematical numeral systems the radix r is usually the number of unique digits, including zero, that a positional numeral system uses to represent numbers.
- The highest symbol of a positional numeral system usually has the value one less than the value of the radix of that numeral system. The standard positional numeral systems differ from one another only in the base they use.
- The radix is an integer that is greater than 1, since a radix of zero would not have any digits, and a radix of 1 would only have the zero digit.

Binary numeral system base

$$A = \sum_{i=0}^{n-1} R^i \cdot d_i$$

$$\overset{2}{1}\overset{1}{2}\overset{0}{3}_{(10)} = (10^2 \cdot 1) + (10^1 \cdot 2) + (10^0 \cdot 3)$$

$$\overset{7}{0}\overset{6}{1}\overset{5}{1}\overset{4}{1}\overset{3}{1}\overset{2}{0}\overset{1}{1}\overset{0}{1}_{(2)} = (2^7 \cdot 0) + (2^6 \cdot 1) + (2^5 \cdot 1) + (2^4 \cdot 1) + (2^3 \cdot 1) + (2^2 \cdot 0) + (2^1 \cdot 1) + (2^0 \cdot 1) = 123_{(10)}$$

Converting decimal numbers to binary

Steps:

1. Divide the decimal number by 2. The remainder of the division will be either 0 or 1.
2. If the remainder is 0, write down a 0.
3. If the remainder is 1, write down a 1.
4. Repeat steps 2 and 3 until the decimal number is 0.
5. Read the numbers **from bottom to top** to get the binary number.

Example

123_{10}

- $123 / 2 = 61$ (remainder 1)
- Write down a 1.
- $61 / 2 = 30$ (remainder 1)
- Write down a 1.
- $30 / 2 = 15$ (remainder 0)
- Write down a 0.
- $15 / 2 = 7$ (remainder 1)
- Write down a 1.
- $7 / 2 = 3$ (remainder 1)
- Write down a 1.

Converting binary numbers to oct

1111011₂

Radix	Value	Binary Value	Symbol
Octal	0	0000	0
Octal	1	0001	1
Octal	2	0010	2
Octal	3	0011	3
Octal	4	0100	4
Octal	5	0101	5
Octal	6	0110	6
Octal	7	0111	7
Hexadecimal	8	1000	8
Hexadecimal	9	1001	9
Hexadecimal	10	1010	A
Hexadecimal	11	1011	B
Hexadecimal	12	1100	C
Hexadecimal	13	1101	D
Hexadecimal	14	1110	E
Hexadecimal	15	1111	F

001.111.011 = 1.7.3

0111.1011 = 7.B

Unsigned Representation

 USR

Sign-Magnitude Representation

 SM

Data in a computer

can essentially be stored using two standards:

1. integers represented in binary,
2. real (floating-point) numbers stored according to the IEEE 754 standard.

Everything else is a combination or interpretation based on these two fundamental forms of representation.

Summary

This archive provides a pre-configured, lightweight MinGW (Minimalist GNU for Windows) environment tailored for compiling and building C and C++ programs directly from the command line or via Makefile scripts.

What is MinGW?

MinGW is a minimalist development environment for native Microsoft Windows applications. It provides a complete open-source programming toolchain that allows you to use standard GNU utilities on Windows.

Included Toolchain (32-bit)

GCC / G++: The GNU C and C++ compilers required to translate your source code into executable binaries.

GNU Make: Build automation tool used to execute Makefile instructions and manage

Summary

Environment Setup & Build Basics

Welcome to the starter project! This is a minimalist C project designed to help you get familiar with Visual Studio Code, manual compilation, and basic build tools.

Project Structure

```
├── compile.bat      # Quick script for manual command-line compilation
├── main.c           # Simple starting C source code
├── Makefile         # Standard build automation script
└── .vscode
    ├── launch.json  # Debugger configuration
    ├── settings.json # Editor workspace settings
    └── tasks.json    # Build tasks integration for VS Code
```

Prerequisites (Required)

Before opening this project, ensure you have the following installed on your system: